Question	Answer	Marks	Partial Marks
8(a)	$\frac{12}{x}+\frac{26}{x+10}=2.8$ oe isw	3	**B2** for $\frac{12}{x}+\frac{26}{x+10}$ oe isw OR **B1** for $\frac{26}{x+10}$ seen **B1** for time = 2.8 or $\frac{168}{60}$ or $2\frac{48}{60}$ oe
8(b)	$12(x+10)+26x=2.8x(x+10)$ or better	M2	**FT** *their* time, provided 2 algebraic fractions one in x and other in $\pm x \pm 10$ **M1** for $12(x+10)+26x$ seen or better
	$12x+120+26x=2.8x^2+28x$	M1	**FT** *their* equation dep on M2
	$2.8x^2-10x-120=0$ oe or $30x + 300 + 65x = 7x^2 + 70x$ or better leading to $7x^2-25x-300=0$	A1	with no errors or omissions
8(c)	$\frac{[--]25\pm\sqrt{([-]25)^2-4\times7\times-300}}{2\times7}$ oe	B2	**B1** for $\sqrt{([-]25)^2-4(7)(-300)}$ or better or for $\frac{[--]25+\sqrt{q}}{2\times7}$ or $\frac{[--]25-\sqrt{q}}{2\times7}$
	− 5 and 8.57 or 8.571…	B2	**B1** for each or **SC1** for final answers 5 and –8.57
8(d)	84 to 84.01…	2	**FT** $\frac{720}{\textit{their}\text{ positive answer}}$ to 3 sf or better **M1** for $\frac{12}{\textit{their}\text{ positive answer}}$ [× 60] oe
9(a)	54[.0] or 53.99 to 54.03…	6	**M2** for [h =] 95.4 × 3 ÷ (π × 2.4^2) oe or **M1** for $95.4=\frac{1}{3}\times\pi\times2.4^2\times h$ **M2** for [slant ht , l =] $\sqrt{(\textit{their } h)^2+2.4^2}$ or **M1** for $(\textit{their } h)^2 + 2.4^2$ **M1** for $\frac{x}{360}\times2\times\pi\times\textit{their } l=2\times\pi\times2.4$ oe or $\frac{x}{360}\times\pi\times(\textit{their } l)^2=\pi\times2.4\times\textit{their } l$

Question	Answer	Marks	Partial Marks
9(b)	14500 or 14470 to 14480	**4**	**M3** for $200 \times 60 \times 24 \times \pi \times 4^2$ [÷1000] or $2 \times 60 \times 24 \times \pi \times 0.04^2$ [×1000] or **M2** for $200 \times \pi \times 4^2$ or for $2 \times \pi \times 0.04^2$ or **M1** for $\pi \times 4^2$ oe or $\pi \times 0.04^2$ seen oe isw or 1000 cm^3 = 1 litre soi or 1 m^3 = 1000 litres soi or for 24 × 60 seen oe
10(a)	$x^3 + 2x^2 - 5x - 6$ final answer	**3**	**B2** for correct expansion of three brackets unsimplified or for simplified expression of correct form with 3 out of 4 terms correct or **B1** for correct expansion of 2 of the 3 given brackets with at least 3 terms out of four correct
10(b)	$\frac{Mc}{M-2f}$ or $\frac{-Mc}{2f-M}$ final answer	**4**	**M1** for clearing $g - c$ from denominator e.g. $M(g - c) = 2fg$ **M1** for correctly isolating terms in g in numerator on one side **M1** for correctly factorising or simplifying, to single term in g in an equation **M1** for correctly dividing by bracket to final answer
10(c)	$\frac{4x}{x+4}$ final answer	**3**	**B1** for $4x(x - 4)$ **B1** for $(x + 4)(x - 4)$
11(a)(i)	$\frac{1}{6}$ oe on all late branches $\frac{5}{6}$ oe on all not late branches	**2**	**B1** for one correct vertical pair $\frac{1}{6}$ oe and $\frac{5}{6}$ oe
11(a)(ii)	$\frac{5}{36}$ oe	**2**	FT *their* tree **M1** for $\textit{their}\,\frac{1}{6} \times \textit{their}\,\frac{5}{6}$
11(b)(i)	$(G \cup T \cup M)'$ oe	**1**	
11(b)(ii)	28	**1**	
11(b)(iii)	$\frac{17}{50}$ oe	**1**	